

Low diastolic blood pressure and adverse outcomes in inpatients with acute exacerbation of chronic obstructive pulmonary disease: A multicenter cohort study

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Abstract

Background: Although intensively studied in patients with cardiovascular diseases (CVDs), the prognostic value of diastolic blood pressure (DBP) has little been elucidated in patients with acute exacerbation of chronic obstructive pulmonary disease (AECOPD). This study aimed to reveal the prognostic value of DBP in AECOPD patients.

Methods: Inpatients with AECOPD were prospectively enrolled from 10 medical centers in China between September 2017 and July 2021. DBP was measured on admission. The primary outcome was all-cause in-hospital mortality; invasive mechanical ventilation and intensive care unit (ICU) admission were secondary outcomes. Least absolute shrinkage and selection operator (LASSO) and multivariable Cox regressions were used to identify independent prognostic factors and calculate the hazard ratio (HR) and 95% confidence interval (CI) for adverse outcomes.

Results: Among 13,633 included patients with AECOPD, 197 (1.45%) died during their hospital stay. Multivariable Cox regression analysis showed that low DBP on admission (<70 mmHg) was associated with increased risk of in-hospital mortality (HR = 2.16, 95% CI: 1.53–3.05, $Z = 4.37$, $P < 0.01$), invasive mechanical ventilation (HR = 1.65, 95% CI: 1.32–2.05, $Z = 19.67$, $P < 0.01$), and ICU admission (HR = 1.45, 95% CI: 1.24–1.69, $Z = 22.08$, $P < 0.01$) in the overall cohort. Similar findings were observed in subgroups with or without CVDs, except for invasive mechanical ventilation in the subgroup with CVDs. When DBP was further categorized in 5-mmHg increments from <50 mmHg to ≥ 100 mmHg, and 75 to <80 mmHg was taken as reference, HRs for in-hospital mortality increased almost linearly with decreased DBP in the overall cohort and subgroups of patients with CVDs; higher DBP was not associated with the risk of in-hospital mortality.

Conclusion: Low on-admission DBP, particularly <70 mmHg, was associated with an increased risk of adverse outcomes among inpatients with AECOPD, with or without CVDs, which may serve as a convenient predictor of poor prognosis in these patients.

Clinical Trial Registration: Chinese Clinical Trial Registry, No. ChiCTR2100044625.

Keywords: Acute exacerbation of chronic obstructive pulmonary disease; Diastolic blood pressure; Prognostic factors; Inpatients; In-hospital mortality

Chen Zhou and Qun Yi contributed equally to this work.

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Introduction

Chronic obstructive pulmonary disease (COPD) is a leading cause of morbidity and mortality worldwide that causes a substantial and increasing economic and social burden.^[1] The global prevalence of COPD is estimated to be approximately 11.7%, with approximately three million deaths annually.^[2] Acute exacerbation of COPD (AECOPD), a main cause of death in patients with COPD, always involves severe health conditions, a change in regular medication, and even adverse outcomes such as in-hospital mortality, invasive mechanical ventilation, and ICU admission.^[3] With increased severity, the in-hospital mortality of AECOPD varies from 0.1% to 24.5%.^[4–7] As a result, assessment of the severity and outcome prediction of patients with AECOPD at admission are vital in planning an appropriate treatment strategy.

Many researchers have studied the prognostic factors of adverse outcomes, which mainly included intensive care unit (ICU) admission, longer hospital stays, and short- or long-term mortality, in inpatients with AECOPD. Older age, laboratory tests indicating infection or acidosis, and cardiovascular and pulmonary comorbidities or severe dyspnea are known as predictive risk factors for adverse outcomes.^[8–13] Nevertheless, the prognostic value of on-admission vital signs that can be easily obtained from inpatients with AECOPD, including blood pressure, radial pulse, pulse pressure, and respiratory rate, has rarely been studied.

Blood pressure management is an important issue among clinicians. Systolic blood pressure (SBP) garners greater attention than diastolic blood pressure (DBP), which is often overlooked in clinical practice. SBP is mostly discussed in natural populations or patients with cardiovascular disease (CVD),^[14–16] but according to the literature, DBP is more frequently associated with prognosis in CVD and other diseases, and this association could be independent of SBP.^[17–19] DBP has been acknowledged as an explicit and independent risk factor for the development of CVD in high-risk patients, although controversy remains.^[20–22] In most studies, a J-curve or U-shaped relationship has been found between DBP and the risk of CVD occurrence in patients with a high risk of CVD, sometimes even in people with good health; that is, DBP >80 mmHg or DBP <60 mmHg usually predicts a greater risk of CVD.^[21,23–26] High DBP alone has been associated with better outcomes in patients with congenital heart defects.^[27] The association of DBP with mortality in patients with or at risk for CVD has also been intensively studied, although few conclusions have been reached. Some studies show a U-shaped relationship between mortality risk and DBP in some populations, including patients with COPD;^[18,28–30] however, other studies have found that a similar relationship does not exist in patients with or without CVD.^[24,31–35] Nevertheless, all the aforementioned studies indicate the vital prognostic function of low DBP for all-cause mortality among populations with or at increased risk of CVD.

However, the role of DBP in predicting the outcomes of inpatients with AECOPD has not been reported. Thus,

this prospective multicenter study primarily aimed to evaluate the relationship between DBP on admission and in-hospital mortality, invasive mechanical ventilation, and ICU admission in patients hospitalized for AECOPD, with and without CVD.

Methods

Ethical approval

The Ethics Committee of each participating academic medical center approved the study based on the ethical approval obtained from the Ethics Committee of the coordinating center, West China Hospital of Sichuan University (No: 2019-1056). All participants provided written informed consent.

Patients and study design

Patient inclusion was based on the prospective, non-interventional, multicenter cohort MAGNET AECOPD (MAnagement aNd advErse ouTcomes in inpatients with acute exacerbation of COPD), a registry study in China (ChiCTR2100044625). The main aims of this registry study were to investigate the adverse outcomes (including intensive treatment usage; in-hospital venous thromboembolism; short- and long-term mortality; and readmission) of inpatients with AECOPD and to establish and validate early warning models of these adverse outcomes.^[36] The original study enrolled consecutive adult inpatients diagnosed with AECOPD among 10 tertiary hospitals in China between September 2017 and July 2021. The diagnosis of AECOPD was based on the following criteria: (1) a history of COPD, including exposure to risk factors (e.g., tobacco smoking, specific environmental exposure); long-term dyspnea, chronic cough, or sputum production; and post-bronchodilator spirometry testing showing a forced expiratory volume in 1 s/forced vital capacity ratio <70%; and (2) an acute worsening of respiratory symptoms resulting in additional therapy. All the patients enrolled in the MAGNET AECOPD study were included in this analysis except for the patients with missing records of blood pressure on admission. Patients were managed at the discretion of their treating physicians.

Data collection

All physicians in charge of data collection were well trained. A standardized case report form was completed for every enrolled patient, including baseline demographics, comorbidities, vital signs on admission, laboratory results, and treatments. Blood pressure assessment was based on routine measurement in the medical center where the patient received treatment, and was completed within 2 h of admission using a wrist-type or brachial automatic sphygmomanometer, or mercurial sphygmomanometer. The values of SBP and DBP obtained by these methods were averaged over two measurements. For some patients admitted to the ICU, the first blood pressure records from invasive blood pressure monitoring were used. Patients were defined as having CVD

if they had any of the following conditions: coronary heart disease, heart failure, arrhythmia, heart valve disease, or stroke, according to the American Heart Association.

Study outcomes

The primary outcome was all-cause in-hospital mortality. The secondary outcomes were invasive mechanical ventilation and ICU admission during the hospital stay.

Statistical analysis

All collected data were analyzed using R software version 4.1.3 (The R Project for Statistical Computing, Vienna, Austria) and IBM SPSS version 25.0 (IBM Corp., Armonk, NY, USA), and a P -value <0.05 was used to indicate statistically significant differences. Eligible patients were divided into a group with DBP <70 mmHg and a group with DBP ≥ 70 mmHg (based on most studies investigating the association between DBP and adverse outcomes in patients with CVD or at risk of CVD).^[29,31,34,37] Baseline demographics, comorbidities, vital signs on admission, laboratory results, and outcome variables were compared between the two groups. Categorical data are expressed as number with percentage, and the chi-squared test was used for comparisons between groups. Normally distributed data are expressed as mean $\pm$ standard deviation, with Student's t -test used for comparisons; non-normally distributed data are expressed as median (Q_1 , Q_3) and a non-parametric test (Mann-Whitney U test) was applied for comparisons between the two groups.

To explore the independent prognostic factors of in-hospital mortality in the overall cohort, univariable Cox regression was performed and factors with a P -value <0.05 were included in least absolute shrinkage and selection operator (LASSO) regression analysis, which was applied to filter and shrink appropriate variables using the "glmnet" R package to minimize the risk of overfitting. The penalty parameter (λ) for the model was determined in 100-fold cross-validation following the minimum criteria (i.e., the value of λ corresponding to the highest partial likelihood deviance, which is referred to as λ_{1se} in the following description). The chosen variables in LASSO regression were further included in multivariable Cox regression to determine the independent prognostic factors associated with in-hospital mortality. A forest plot was drawn to show the variables included in the prognostic model based on this multivariable Cox regression (individuals with missing values of the included variables were excluded). We also conducted sensitivity analyses, excluding patients with pneumonia or sepsis. Similar strategies of statistical analysis were used for the secondary outcomes (invasive mechanical ventilation and ICU admission).

In addition to binary classification of DBP as <70 mmHg or ≥ 70 mmHg, DBP on admission was further categorized in 5-mmHg increments from <50 mmHg to ≥ 100 mmHg to identify the trend in the risk of adverse outcomes with a change in DBP in the overall cohort and the subgroups

with or without CVD. According to the independent prognostic factors identified in the above analysis, multivariable Cox analysis was performed again to evaluate the risk of in-hospital mortality for different DBP categories. According to multivariable Cox analysis, the adjusted hazard ratio (HR) and 95% confidence interval (CI) for each category of DBP were calculated with reference to the DBP 75 to <80 mmHg, where the HR was considered 1.

In survival analysis, Kaplan–Meier analysis and the log-rank test were used to describe and compare mortality between patients with DBP <70 mmHg and DBP ≥ 70 mmHg in the overall population and in subgroups with or without CVD.

Results

Characteristics of the study population

A total of 14,007 patients admitted for AECOPD were enrolled in the original registration study. In the present analysis, we excluded patients with missing DBP records ($n = 374$), and a total of 13,633 inpatients with AECOPD were included [Supplementary Figure 1, <http://links.lww.com/CM9/B513>]. Among them, 197 (1.45%) died during hospitalization, 453 (3.32%) required invasive mechanical ventilation during their hospital stay, and 1021 (7.49%) patients were admitted to the ICU. A total of 3794 patients (27.83%) had CVD comorbidity.

Characteristics of the included patients with AECOPD are described in Table 1, among which 3108 patients (22.80%) had DBP <70 mmHg on admission and 10,525 patients (77.20%) had DBP ≥ 70 mmHg. As expected, DBP <70 mmHg group patients tended to be older (74.42 ± 10.44 years *vs.* 71.76 ± 10.27 years, $t = -12.87$, $P < 0.001$), with a higher proportion of female patients (24.68% [767/3108] *vs.* 20.49% [2157/10,525], $\chi^2 = 24.93$, $P < 0.001$) than DBP ≥ 70 mmHg group. Additionally, there were significant differences in the prevalence of CVD, especially coronary heart disease (13.90% [432/3108] *vs.* 10.27% [1081/10,525], $\chi^2 = 32.03$, $P < 0.001$), heart failure (15.22% [473/3108] *vs.* 10.31% [1085/10,525], $\chi^2 = 57.15$, $P < 0.001$), arrhythmia (11.49% [357/3108] *vs.* 7.76% [817/10,525], $\chi^2 = 42.28$, $P < 0.001$), and stroke (7.85% [244/3108] *vs.* 5.84% [615/10,525], $\chi^2 = 16.38$, $P < 0.001$), which were more frequently observed in patients with DBP <70 mmHg than in those DBP ≥ 70 mmHg. Other than CVDs comorbidities, bronchiectasis (19.50% [606/3108] *vs.* 17.12% [1802/10,525], $\chi^2 = 9.32$, $P = 0.002$), interstitial lung disease (9.30% [289/3108] *vs.* 8.10% [853/10,525], $\chi^2 = 4.46$, $P = 0.035$), chronic pulmonary heart disease (23.33% [725/3108] *vs.* 20.00% [2105/10,525], $\chi^2 = 16.15$, $P < 0.001$), pneumonia (32.40% [1007/3108] *vs.* 23.81% [2506/10,525], $\chi^2 = 92.57$, $P < 0.001$), sepsis (0.68% [21/3108] *vs.* 0.29% [31/10,525], $\chi^2 = 9.17$, $P = 0.002$), reduced mobility (15.99% [497/3108] *vs.* 12.19% [1283/10,525], $\chi^2 = 30.54$, $P < 0.001$), diabetes (15.54% [483/3108] *vs.* 12.26% [1290/10,525], $\chi^2 = 22.87$, $P < 0.001$), chronic renal failure (5.57% [173/3108] *vs.* 3.15% [332/10,525], $\chi^2 = 39.13$, $P < 0.001$), and anxiety or depression (2.25% [70/3108] *vs.* 1.53% [161/10,525], $\chi^2 = 7.52$, $P = 0.006$)

Table 1: Basic characteristics of the included AECOPD patients.

Characteristics	Total (N = 13,633)	DBP < 70 mmHg (N = 3108)	DBP ≥ 70 mmHg (N = 10,525)	Statistical values	P-value
Baseline characteristics					
Age (years)	72.36 ± 10.37	74.42 ± 10.44	71.76 ± 10.27	-12.87 [†]	<0.001
Female	2924 (21.45)	767 (24.68)	2157 (20.49)	24.93 ^{**}	<0.001
Current smoking	2735 (20.06)	604 (19.43)	2131 (20.25)	0.99 ^{**}	0.320
BMI (kg/m ²)*	21.3 (18.71–24.03)	20.89 (18.29–23.53)	21.48 (18.83–24.22)	-5.56 ^{††}	<0.001
BMI ≥ 30 kg/m ² *	206 (2.77)	45 (2.41)	161 (2.89)	1.18 ^{**}	0.277
Comorbidities					
CVDs	3794 (27.83)	1046 (33.66)	2748 (26.11)	68.02 ^{**}	<0.001
Coronary heart disease	1513 (11.10)	432 (13.90)	1081 (10.27)	32.03 ^{**}	<0.001
Heart failure	1558 (11.43)	473 (15.22)	1085 (10.31)	57.15 ^{**}	<0.001
Heart valve diseases	248 (1.82)	68 (2.19)	180 (1.71)	3.07 ^{**}	0.080
Arrhythmia	1174 (8.61)	357 (11.49)	817 (7.76)	42.28 ^{**}	<0.001
Stroke	859 (6.30)	244 (7.85)	615 (5.84)	16.38 ^{**}	<0.001
Hypertension	4579 (33.59)	1016 (32.69)	3563 (33.85)	1.46 ^{**}	0.228
Pulmonary tuberculosis	621 (4.56)	157 (5.05)	464 (4.41)	2.28 ^{**}	0.131
Bronchiectasis	2408 (17.66)	606 (19.50)	1802 (17.12)	9.32 ^{**}	0.002
ILD	1142 (8.38)	289 (9.30)	853 (8.10)	4.46 ^{**}	0.035
Asthma	456 (3.34)	97 (3.12)	359 (3.41)	0.62 ^{**}	0.430
OSAHS	85 (0.62)	15 (0.48)	70 (0.67)	1.29 ^{**}	0.256
Chronic pulmonary heart disease	2830 (20.76)	725 (23.33)	2105 (20.00)	16.15 ^{**}	<0.001
Pneumonia	3513 (25.77)	1007 (32.40)	2506 (23.81)	92.57 ^{**}	<0.001
Active cancer	727 (5.33)	184 (5.92)	543 (5.16)	2.75 ^{**}	0.097
Sepsis	52 (0.38)	21 (0.68)	31 (0.29)	9.17 ^{**}	0.002
Reduced mobility	1780 (13.06)	497 (15.99)	1283 (12.19)	30.54 ^{**}	<0.001
Diabetes	1773 (13.01)	483 (15.54)	1290 (12.26)	22.87 ^{**}	<0.001
Chronic renal failure	505 (3.70)	173 (5.57)	332 (3.15)	39.13 ^{**}	<0.001
GERD	131 (0.96)	38 (1.22)	93 (0.88)	2.90 ^{**}	0.089
Anxiety–depression	231 (1.69)	70 (2.25)	161 (1.53)	7.52 ^{**}	0.006
Osteoporosis	341 (2.50)	92 (2.96)	249 (2.37)	3.48 ^{**}	0.062
Vital Signs					
Systolic blood pressure (mmHg)	130 (120–145)	117 (105–129)	134 (125–147)	-44.54 ^{††}	<0.001
Diastolic blood pressure (mmHg)	79 (70–87)	64 (59–67)	83 (76–89)	-84.88 ^{††}	<0.001
Mean arterial pressure (mmHg)	97 (88–105)	82 (76–86)	100 (94–107)	-76.41 ^{††}	<0.001
Pulse pressure (mmHg)	52 (40–63)	54 (44–66)	51 (40–63)	-9.51 ^{††}	<0.001
Radial pulse (/min) [‡]	88 (79–100)	86 (77–98)	88 (80–100)	-6.60 ^{††}	<0.001
Respiratory rate (/min) [‡]	20 (20–22)	20 (20–22)	20 (20–22)	-2.64 ^{††}	0.008
Laboratory tests					
Red blood cell (×10 ¹² /L) [§]	4.29 (3.82–4.73)	4.10 (3.62–4.58)	4.34 (3.89–4.77)	-15.29 ^{††}	<0.001
Hemoglobin (g/L) [§]	129 (115–141)	123 (108–137)	130 (117–142)	-16.00 ^{††}	<0.001
White blood cell (×10 ⁹ /L) [§]	7.70 (5.90–10.18)	7.78 (5.76–10.40)	7.69 (5.90–10.10)	-0.36 ^{††}	0.716
NEUT (%) [§]	75.20 (65.70–84.30)	75.90 (66.10–84.80)	75.00 (65.60–84.20)	-2.02 ^{††}	0.040
EOSR (%) [§]	1.00 (0.10–2.60)	0.90 (0.10–2.50)	1.00 (0.10–2.60)	-1.68 ^{††}	0.094
Platelet (×10 ⁹ /L) [§]	192 (145–249)	189 (139–248)	192 (146–249)	-2.33 ^{††}	0.020
pH	7.41 (7.37–7.44)	7.41 (7.37–7.44)	7.41 (7.37–7.44)	-1.72 ^{††}	0.086
PaO ₂ (mmHg)	82.90 (68.1–106.0)	83.70 (68.30–108.00)	82.50 (68.10–105.40)	-2.00 ^{††}	0.046
PaCO ₂ (mmHg)	42.70 (36.90–51.60)	42.30 (36.40–51.60)	42.80 (37.00–51.60)	-1.31 ^{††}	0.190
Lactate (mmol/L)	1.50 (1.20–1.90)	1.50 (1.20–1.90)	1.50 (1.20–1.90)	-1.27 ^{††}	0.203
Adverse Outcomes					
In-hospital mortality	197 (1.45)	104 (3.35)	93 (0.88)	102.18 ^{**}	<0.001
Invasive mechanical ventilation	453 (3.32)	194 (6.24)	259 (2.46)	106.79 ^{**}	<0.001
ICU admission	1021 (7.49)	352 (11.35)	669 (6.36)	85.52 ^{**}	<0.001
Length of Stay (days)	9 (7–14)	9 (7–14)	9 (7–13)	-4.32 ^{††}	<0.001

Data are presented as the number (%), mean ± standard deviation, or median (Q₁, Q₃). * Available in 7435 patients; † available in 13,600 patients; ‡ available in 13,589 patients; § available in 11,425 patients; || available in 8879 patients. †t values; **χ² values; ††U values. AECOPD: Acute exacerbation of chronic obstructive pulmonary disease; BMI: Body mass index; CVDs: Cardiovascular diseases; DBP: Diastolic blood pressure; EOSR: Eosinophil percentage; GERD: Gastroesophageal reflux disease; ICU: Intensive care unit; ILD: Interstitial lung disease; NEUT: Neutrophil percentage; OSAHS: Obstructive sleep apnea-hypopnea syndrome; PaCO₂: Partial pressure of carbon dioxide; PaO₂: Partial pressure of oxygen.

were more frequently in patients with DBP <70 mmHg compared with those with DBP ≥70 mmHg. Regarding vital signs on admission, patients with DBP <70 mmHg tended to have lower SBP (117 [105, 129] mmHg *vs.* 134 [125, 147] mmHg, $U = -44.54$, $P < 0.001$) and slower radial pulse (86 [77, 98] *vs.* 88 [80, 100], $U = -6.60$, $P < 0.001$) as well as higher pulse pressure (54 [44, 66] mmHg *vs.* 51 [40, 63] mmHg, $U = -9.51$, $P < 0.001$) than patients with DBP ≥70 mmHg. In terms of laboratory tests, patients with DBP <70 mmHg had lower red blood cell counts (4.10 [3.62, 4.58] $\times 10^{12}/L$ *vs.* 4.34 [3.89, 4.77] $\times 10^{12}/L$, $U = -15.29$, $P < 0.001$), hemoglobin (Hb) (123 [108, 137] g/L *vs.* 130 [117, 142] g/L, $U = -16.00$, $P < 0.001$), and platelet counts (189 [139, 248] $\times 10^9/L$ *vs.* 192 [146, 249] $\times 10^9/L$, $U = -2.33$, $P = 0.020$), and higher neutrophil percentage (NEUT%) (75.90% [66.10%, 84.80%] *vs.* 75.00% [65.60%, 84.20%], $U = -2.02$, $P = 0.040$) as well as slightly higher partial pressure of oxygen in arterial blood (83.70 [68.30, 108.00] mmHg *vs.* 82.50 [68.10, 105.40] mmHg, $U = -2.00$, $P = 0.046$) than patients with DBP ≥70 mmHg. Compared with patients with DBP ≥70 mmHg, the rates of in-hospital mortality (3.35% [104/3108] *vs.* 0.88% [93/10,525], $\chi^2 = 102.18$, $P < 0.001$), invasive mechanical ventilation (6.24% [194/3108] *vs.* 2.46% [259/10,525], $\chi^2 = 106.79$, $P < 0.001$), and ICU admission (11.35% [352/3108] *vs.* 6.36% [669/10,525], $\chi^2 = 85.52$, $P < 0.001$) were significantly higher in patients with DBP <70 mmHg, and the length of hospital stay was significantly longer (9 [7, 14] days *vs.* 9 [7, 13] days, $U = -4.32$, $P < 0.001$) in patients with DBP <70 mmHg.

Factors associated with in-hospital mortality in the overall cohort

A univariable Cox analysis was conducted based on the overall cohort to explore the factors related to in-hospital mortality [Supplementary Table 1, <http://links.lww.com/CM9/B513>], and those variables with P -value <0.05 were included in LASSO regression. After λ_{1se} was determined [Supplementary Figure 2, <http://links.lww.com/CM9/B513>], the selected variables were further included in multivariable Cox regression analysis. A forest plot was drawn according to multivariable Cox regression [Figure 1], in which DBP <70 mmHg was an independent risk factor for in-hospital mortality (HR = 2.16, 95% CI = [1.53–3.05], $P < 0.001$), after adjusting for other prognostic factors. Other independent risk factors included heart failure (HR = 1.51, 95% CI = [1.10–2.09], $P = 0.012$), chronic pulmonary heart disease (HR = 1.72, 95% CI = [1.25–2.36], $P < 0.001$), pneumonia (HR = 2.28, 95% CI = [1.63–3.17], $P < 0.001$), interstitial lung disease (HR = 1.76, 95% CI = [1.15–2.69], $P = 0.009$), sepsis (HR = 1.95, 95% CI = [1.03–3.69], $P = 0.041$), reduced mobility (HR = 1.79, 95% CI = [1.29–2.48], $P < 0.001$), older age (HR = 1.41, 95% CI = [1.20–1.66], $P < 0.001$), higher radial pulse (HR = 1.19, 95% CI = [1.05–1.36], $P = 0.008$), and higher NEUT% (HR = 1.39, 95% CI = [1.13–1.71], $P = 0.002$), and lower Hb value (HR = 0.83, 95% CI = [0.72–0.96], $P = 0.011$). Chronic renal failure, SBP, and white blood cell count were not independent risk factors for in-hospital mortality. Sensitivity analysis, excluding patients with pneumonia or sepsis, also revealed similar results [Supplementary Figure 3, <http://links.lww.com/CM9/B513>].

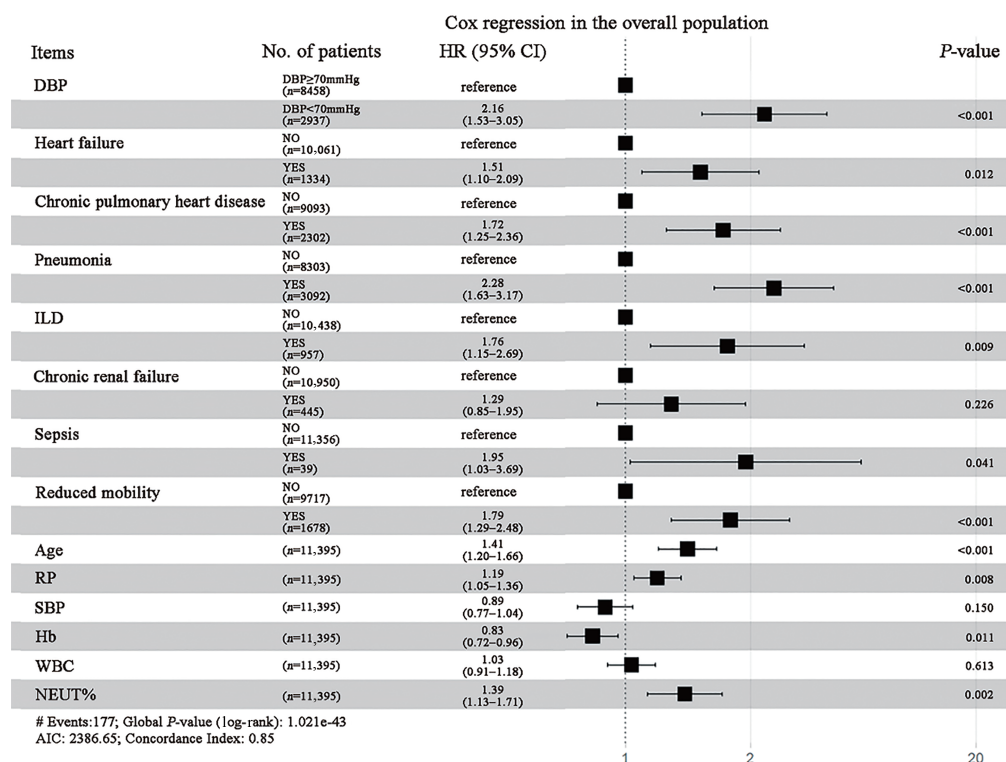


Figure 1: Forest plot showing HRs for the risk factors of in-hospital mortality based on multivariable Cox regression in the overall population. AIC: Akaike information criterion; CI: Confidence interval; DBP: Diastolic blood pressure; Hb: Hemoglobin; HRs: Hazard ratios; ILD: Interstitial lung disease; NEUT%: Neutrophil percentage; RP: Radial pulse; SBP: Systolic blood pressure; WBC: White blood cell.

DBP and adverse outcomes in the overall cohort and subgroups with or without CVD

Table 2 shows the risks for in-hospital mortality, invasive mechanical ventilation, and ICU admission in association with DBP in the overall cohort. Low on-admission DBP (<70 mmHg) was associated with increased risk for in-hospital mortality (HR = 2.16, 95% CI: 1.53–3.05, $P < 0.001$), invasive mechanical ventilation (HR = 1.65, 95% CI: 1.32–2.05, $P < 0.001$), and ICU admission (HR = 1.45, 95% CI: 1.24–1.69, $P < 0.001$) in the overall cohort, after adjustment of other independent prognostic factors. Furthermore, when the DBP was categorized in 5-mmHg increments from <50 mmHg to ≥ 100 mmHg, and 75 to <80 mmHg

was taken as the reference, there was evident increased risk of above adverse outcomes with the decreased DBP, while these trends were not observed with increased DBP [Table 2].

Similar findings were observed in subgroups of patients with or without CVD [Table 3]. The HR of in-hospital mortality was 2.51 (95% CI: 1.74–3.60, $P < 0.001$) in patients with CVD and 2.26 (95% CI: 1.40–3.65, $P = 0.001$) in patients without CVD for DBP <70 mmHg. DBP <70 mmHg also increased the risk of most secondary adverse outcomes in subgroups with or without CVD, with only one exception of the invasive mechanical ventilation in patients with CVD.

Table 2: Associations of DBP with adverse outcomes in the overall AECOPD inpatients by multivariable Cox regression.

Variables	In-hospital mortality		Invasive mechanical ventilation		ICU admission	
	HR (95% CI)	P-value*	HR (95% CI)	P-value*	HR (95% CI)	P-value*
Binary classification of DBP (mmHg)						
DBP ≥ 70	Reference		Reference		Reference	
DBP <70	2.16 (1.53–3.05)	<0.001	1.65 (1.32–2.05)	<0.001	1.45 (1.24–1.69)	<0.001
Multiclass classification of DBP (mmHg)						
<50	6.26 (3.11–12.59)	<0.001	4.83 (2.78–8.39)	<0.001	2.67 (1.67–4.26)	<0.001
50 to <55	4.59 (2.19–9.65)	<0.001	2.18 (1.18–4.05)	0.014	1.90 (1.21–2.98)	0.006
55 to <60	5.16 (2.66–10.00)	<0.001	1.98 (1.23–3.18)	0.005	1.67 (1.19–2.34)	0.003
60 to <65	2.51 (1.32–4.77)	0.005	1.60 (1.05–2.44)	0.028	1.29 (0.96–1.74)	0.091
65 to <70	2.50 (1.36–4.61)	0.003	1.02 (0.68–1.54)	0.932	1.02 (0.77–1.34)	0.906
70 to <75	1.70 (0.88–3.27)	0.112	0.97 (0.65–1.45)	0.879	0.84 (0.64–1.10)	0.208
75 to <80	Reference		Reference		Reference	
80 to <85	1.84 (0.98–3.45)	0.056	1.01 (0.65–1.55)	0.980	1.00 (0.76–1.31)	0.985
85 to <90	0.96 (0.46–2.01)	0.909	0.75 (0.46–1.22)	0.242	0.82 (0.61–1.11)	0.207
90 to <95	0.74 (0.27–2.04)	0.563	0.79 (0.45–1.38)	0.409	0.82 (0.58–1.17)	0.280
95 to <100	1.07 (0.39–2.95)	0.895	0.86 (0.45–1.65)	0.656	0.91 (0.61–1.37)	0.666
≥ 100	0.90 (0.30–2.70)	0.843	1.86 (1.10–3.13)	0.020	1.00 (0.68–1.49)	0.982

AECOPD: Acute exacerbation of chronic obstructive pulmonary disease; CI: Confidence interval; DBP: Diastolic blood pressure; HR: Hazard ratio; ICU: Intensive care unit. *Adjusted for age, heart failure, chronic pulmonary heart disease, pneumonia, interstitial lung disease, sepsis, reduced mobility, radial pulse, neutrophil percentage, and hemoglobin.

Table 3: Associations of DBP with adverse outcomes in the AECOPD inpatients with or without CVD by multivariable Cox regression.

Variables	In-hospital mortality		Invasive mechanical ventilation		ICU admission	
	HR (95% CI)	P-value	HR (95% CI)	P-value	HR (95% CI)	P-value
With CVDs ($n = 3794$)						
DBP ≥ 70 mmHg	Reference		Reference		Reference	
DBP < 70 mmHg	2.51 (1.74–3.60)	<0.001	0.83 (0.52–1.34)	0.447	1.33 (1.07–1.67)	0.012
Without CVDs ($n = 9839$)						
DBP ≥ 70 mmHg	Reference		Reference		Reference	
DBP < 70 mmHg	2.26 (1.40–3.65)	0.001	2.00 (1.48–2.71)	<0.001	1.49 (1.21–1.83)	<0.001

AECOPD: Acute exacerbation of chronic obstructive pulmonary disease; CI: Confidence interval; CVDs: Cardiovascular diseases; DBP: Diastolic blood pressure; HR: Hazard ratio; ICU: Intensive care unit.

Figure 2 further depicts the association between DBP and the risk of in-hospital mortality in the study population. With reference to DBP 75 to <80 mmHg, the HRs for in-hospital mortality increased almost linearly with decreased DBP in the overall cohort and subgroup of patients with CVD; higher DBP was not associated with a higher or lower risk of in-hospital mortality. The trend remained in the subgroup of patients without CVD, although it was not as evident as those in the overall cohort and in patients with CVD.

Kaplan–Meier survival analysis of the population by DBP level

The predictive value of DBP <70 mmHg for death during the hospital stay was further investigated by time-to-event analysis in the overall cohort as well as the two subgroups with or without CVD [Figure 3]. On-admission DBP (<70 mmHg or ≥70 mmHg) could distinguish the risk of in-hospital mortality in the overall cohort and both subgroups (all $P < 0.05$ by log-rank test).

Discussion

In this study, we investigated the prognostic role of DBP for in-hospital adverse outcomes among inpatients with AECOPD. After adjusting other factors associated with

the primary outcome of this population, DBP <70 mmHg was independently associated with a higher risk of in-hospital mortality in the overall cohort and in subgroups with or without CVD. With decreased DBP (reference to 75 to <80 mmHg), the risk of in-hospital mortality was increased in the overall cohort as well as in the subgroups of patients with or without CVD. Similarly, lower DBP was associated with a higher risk of ICU admission in the overall cohort, and a higher risk of invasive mechanical ventilation in the overall cohort.

Among inpatients with AECOPD enrolled in this study, 197 (1.45%) died during their hospital stay. The in-hospital mortality rate was very close to that reported in another large-scale multicenter study conducted in China, which included 4569 inpatients with AECOPD from 41 hospitals, in which 1.24% of patients died during the hospital stay.^[38] Nevertheless, this rate was lower than some other studies that included Caucasian populations, patients with respiratory failure, and those requiring mechanical ventilation or intensive care.^[4,7,39,40] In summary, the difference regarding in-hospital mortality among studies could be explained by ethnic differences and the different inclusion criteria for study populations. In the present study, we consecutively included unselected inpatients with AECOPD from 10 tertiary

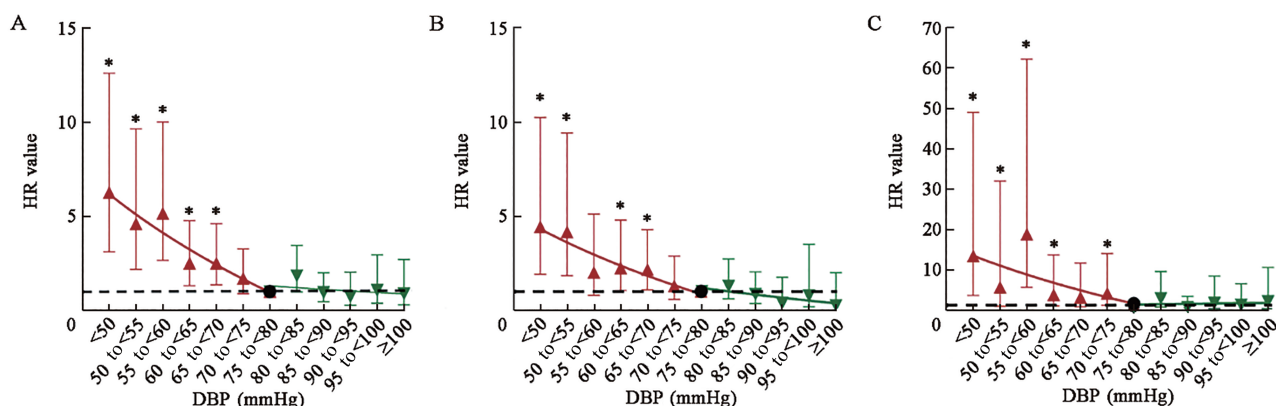


Figure 2: Relationship between DBP (mmHg) and the risk of in-hospital mortality. HR values were obtained by multivariable Cox regression with reference to the DBP group of 75 to <80 mmHg (HR defined as 1) in (A) the overall population; (B) patients with CVDs; and (C) patients without CVDs. The upper and lower 95% CIs are denoted by the vertical solid lines with endpoints. The horizontal dotted line refers to the reference line of HR = 1; the black thick dot refers to the reference (i.e., 75 to <80 mmHg; the red triangle refers to the DBP lower than the reference; the green triangle refers to the DBP higher than the reference). * $P < 0.05$. CIs: Confidence intervals; CVDs: Cardiovascular diseases; DBP: Diastolic blood pressure; HR: Hazard ratio.

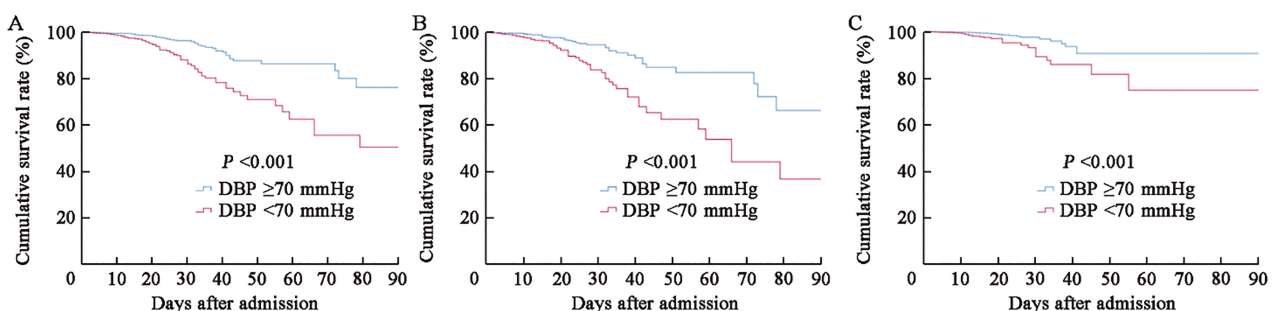


Figure 3: Kaplan–Meier estimates for in-hospital survival in patients with DBP <70 mmHg and DBP ≥70 mmHg in (A) the overall population; (B) patients with CVDs; and (C) patients without CVDs (all $P < 0.05$ by log-rank test). CVDs: Cardiovascular diseases; DBP: Diastolic blood pressure.

general hospitals of China, which should represent real-world mortality among inpatients with AECOPD in China.

Despite the availability of robust data on relationships between DBP and mortality in populations with CVD or at risk for CVD,^[20,27,28,31,32,41] only scant data in the population of patients with COPD are available. High blood pressure, regardless of SBP (>140 mmHg) or DBP (>90 mmHg), seriously endangers the health and even life of individuals, and the pathophysiological mechanism is easy to understand. However, in recent decades, concerns are rising regarding the potentially increased risk of adverse outcomes owing to low DBP in patients with CVD or at risk for CVD. A U-shaped or J-shaped relationship between DBP and adverse outcomes (including all-cause mortality, cardiovascular death, and cardiovascular events) has been reported in many large-scale studies, with progressively increased and decreased values of DBP outside the optimal ranges being associated with increased mortality rates.^[30] For example, Byrd *et al*^[30] found that a “U-shaped” relationship between DBP and all-cause mortality exists in patients with COPD and increased cardiovascular risk; DBP lower than 80 mmHg or higher than 90 mmHg were both significantly associated with a higher risk of death in the follow-up period. In our study, DBP <70 mmHg was significantly associated with in-hospital mortality among inpatients with AECOPD, but the relationship between DBP and the risk of mortality did not present a U-shaped. With reference to 75 to <80 mmHg, the risk of all-cause mortality increased with decreased DBP whereas higher DBP was not associated with higher risk of in-hospital mortality. The trends were consistently observed across overall inpatients with AECOPD and subgroups with CVD and without CVD. Similarly to our results, Vámos *et al*^[35] found that low DBP was independently related to a higher death rate in patients with type 2 diabetes; additionally, in patients with pulmonary embolism, low DBP showed prognostic value for both short- and long-term mortality.^[42] An increased risk of mortality was not found with higher DBP in these study populations. Studies conducted in populations with CVD also found that only low, not high, DBP was associated with increased risk of all-cause mortality.^[31,35,43]

Regarding the underlying mechanism in the association between low DBP and increased mortality risk in populations with CVD or at risk for CVD, varying opinions persist despite decades of research. There are two main explanations for this association. First, myocardial perfusion is critically dependent on DBP because 85% of left ventricular perfusion occurs during diastole.^[44] A lower DBP could potentially lead to myocardial hypoperfusion and was associated myocardial damage. In the Atherosclerosis Risk in Communities cohort study, lower DBP was associated with higher serum concentrations of cardiac troponin T (cTNT), which is a well-known marker of myocardial injury.^[45] Other than coronary perfusion, DBP also indicates the perfusion status of various end organs; adverse outcomes have been found in the eyes, kidneys, and brain when DBP is lower than normal levels.^[19,32,37,46,47] Second, stiff vessels may reduce DBP by impairing the reservoir function of the aorta.

Therefore, a low DBP can be regarded as a marker of aortic stiffness and pre-existing vascular disease. This is supported by findings showing that patients with stiff arteries and presumably low DBP have a lower ischemic threshold than control individuals.^[48] That is, the association of DBP with in-hospital mortality and other adverse outcomes might be owing to increased comorbidities and poor underlying health in these patients with low DBP. Among inpatients with AECOPD in the present study, there is particular concern that hypotension, including low SBP and low DBP, may reflect concomitant CVD, pneumonia, or sepsis/septic shock. However, low SBP was not associated with in-hospital mortality in multivariable Cox regression, while low DBP was an independent prognostic factor of in-hospital mortality after adjusting for confounding factors such as age, comorbidities including heart failure, chronic pulmonary heart disease, pneumonia, and sepsis, as well as laboratory test results. As for SBP, low SBP usually reflects an impaired ejection function, which is mostly the result of coronary dysfunction or primary cardiac organic disease.^[49] This is rarely the leading cause of death in patients with AECOPD. Furthermore, a sensitivity analysis excluding patients with pneumonia or sepsis and a subgroup analysis of patients without CVD all did not alter the association of low DBP with mortality. Thus, these findings indicate a new possibility that low DBP may be a part of the AECOPD process and may indicate the severity of AECOPD. Further studies are warranted to confirm our findings and to clarify the potential mechanism of this association. The data herein clearly show that low DBP is associated with an increased risk of adverse outcomes, especially in-hospital mortality, among unselected AECOPD inpatients and AECOPD patients with or without CVD. These patients with a low on-admission DBP should be managed with special attention; and low DBP should be seriously considered when developing early warning systems of adverse outcomes for inpatients with AECOPD.

This study had several strengths. Although the association between low DBP and adverse outcomes has been extensively studied in the field of CVDs, this was large-scale multicenter study to evaluate the prognostic value of DBP for adverse outcomes among inpatients with AECOPD. The prospective and consecutive inclusion of unselected inpatients with AECOPD in our study ensured high data quality, and comprehensive information, including baseline demographics, comorbidities, and laboratory test results, was included in prognostic analysis. LASSO regression analysis was also performed to prevent overfitting. Therefore, these results reflect true associations between DBP and adverse outcomes among AECOPD patients in the real-world setting. This study also had some limitations. First, because of the observational nature of the study, our findings regarding increased risk of death related to low DBP may not imply causality, and we can only speculate about the potential mechanisms that explain these associations. Second, some laboratory data, such as cTNT and N-Terminal pro-brain natriuretic peptide (NT-pro-BNP) values, were missing in some patients; hence, those variables were not included in the multivariable analysis.

Finally, the lack of follow-up data prevented us from further evaluating the association of DBP with long-time outcome in patients with AECOPD after discharge.

In conclusion, low DBP, particularly <70 mmHg, was associated with an increased risk for in-hospital mortality and other adverse outcomes in AECOPD inpatients, with or without known CVD. Although no causality can be implied for these relationships, these findings can serve to alert health care providers regarding possible worse prognosis among inpatients with AECOPD who have low DBP. Further studies are warranted to confirm our findings and to elucidate the potential mechanism of this association. Further investigation is also needed regarding the correlation between DBP and long-term prognosis (including mortality and readmission) of inpatients with AECOPD.

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Conflicts of interest

None.

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